

Name: Key

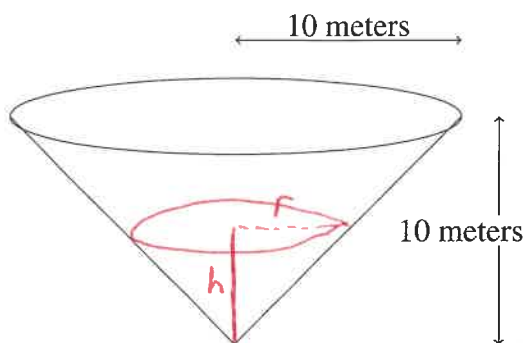
/ 25

Please circle your instructor's name: Kevin Meek James Gossell Margaret Short

There are 25 points possible on this quiz. No aids (book, calculator, etc.) are permitted. **Show all work for full credit.**

1. [9 points] A conical tank is filled with water at a rate of 30 cubic meters per second. The height and radius of the tank are both 10 meters. (See picture below.) How fast is the height of the water changing when the height of the water is 2 meters high? Include units with your answer.

(Recall that the volume of a cone is given by $V = \frac{1}{3}\pi r^2 h$ where V is volume, r is the radius of the top of the cone and h is the height of the cone.)



Note that $\frac{h}{r} = \frac{10}{10} = 1 \Rightarrow h = r$

$$\frac{dV}{dt} = 30 \frac{\text{m}^3}{\text{sec}} \quad \frac{dh}{dt} = ?$$

$$V = \frac{1}{3}\pi h^3 \quad (\text{since } r=h)$$

$$\frac{d}{dt}[V] = \frac{d}{dt}\left[\frac{1}{3}\pi h^3\right]$$

$$\frac{dV}{dt} = \pi h^2 \frac{dh}{dt}$$

$$30 = \pi (2)^2 \frac{dh}{dt}$$

$$\boxed{\frac{dh}{dt} = \frac{15}{2\pi} \frac{\text{m}}{\text{sec}}}$$

2. [8 points] Let $g(x) = \frac{1}{x+1}$.

a. Find the linear approximation of $g(x)$ at $x = 1$.

$$g(1) = \frac{1}{2}, \quad g'(x) = -\frac{1}{(x+1)^2}, \quad g'(1) = -\frac{1}{4}$$

$$L(x) = \frac{1}{2} - \frac{1}{4}(x-1)$$

b. Find the differential of $g(x)$.

$$dy = -\frac{1}{(x+1)^2} dx$$

c. If x changes from $x = 1$ to $x = 1.25$, use the differential to estimate how much you expect $g(x)$ to change. Express your answer as a decimal or as a simplified fraction.

$$\Delta y \approx -\frac{1}{(1+1)^2} (0.25) = -\frac{1}{4} \cdot \frac{1}{4} = -\frac{1}{16}$$

3. [8 points] Let $h(x) = x^2(3-4x)$.

$$\cancel{h(x) = x^2(3-4x)} \quad h(x) = 3x^2 - 4x^3$$

a. Find all critical points of $h(x)$.

$$h'(x) = 6x - 12x^2 = 6x(1-2x)$$

$$x = 0 \quad x = \frac{1}{2}$$

b. Determine the absolute maximum and absolute minimum of $h(x)$ on the interval $[-1, 1]$

x	$h(x)$
-1	7
0	0
$\frac{1}{2}$	$\frac{1}{4}$
1	-1

maximum value of $h(x)$: 7

minimum value of $h(x)$: -1

on $[-1, 1]$